

INTERNSHIP PROJECT REPORT

AI-Powered Resume Screening & Job Recommendation System

Artificial Intelligence Internship

Submitted by

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Organization: Codec Technologies Pvt. Ltd.

Internship Duration: 08/07/2026 – 08/08/2026

Internship Details

Internship Detail	Information
Candidate Name	Aviek Pratap Singh
Roll Number	25SCS1003001203
Degree	B.Tech (CSE)
University	IILM University
Internship	Artificial Intelligence Internship
Organization	Codec Technologies Pvt. Ltd.
Project	AI-Powered Resume Screening & Job Recommendation System
Duration	08/07/2026 – 08/08/2026

1. Executive Summary

This internship report presents the work completed during a one-month Artificial Intelligence Internship at Codec Technologies Pvt. Ltd. The central project was an AI-Powered Resume Screening & Job Recommendation System. The project addressed the challenge of manually reviewing large numbers of resumes and matching candidates with suitable job opportunities.

The proposed workflow processes resume and job-description text, performs text preprocessing, extracts relevant skills and other features, calculates candidate–job similarity, applies matching and ranking logic, and produces ranked job recommendations. The internship provided practical exposure to Python-based data preparation, NLP processing, feature engineering, similarity-based machine learning, and an application layer for interactive recommendations.

2. Introduction

Recruitment teams may receive hundreds of resumes for a single role. Manual screening can be time-consuming and may make it difficult to consistently compare candidates against different role requirements. The internship project explored how Artificial Intelligence and Natural Language Processing can support this process.

The project focuses on converting unstructured resume and job-description text into useful features and then using those features to estimate compatibility. Instead of relying only on exact keyword matches, the workflow described in the internship material uses NLP features and similarity-based matching to support more relevant recommendations.

3. Internship and Candidate Profile

The internship was completed as an Artificial Intelligence Internship at Codec Technologies Pvt. Ltd. in a one-month hybrid format. The candidate profile in the source presentation identifies Aviek Pratap Singh as a B.Tech (CSE) student at IILM University, academic batch 2026–2027.

Internship duration: 08/07/2026 – 08/08/2026

Project: AI-Powered Resume Screening & Job Recommendation System

Organization: Codec Technologies Pvt. Ltd.

4. Organization Profile

Codec Technologies Pvt. Ltd. is presented in the source material as a global platform focused on empowering learners and connecting diverse talent with meaningful career opportunities. The internship emphasized practical learning through project work and training tasks, with a focus on Artificial Intelligence, project development, and practical implementation.

The program format shown in the presentation was a one-month internship conducted in hybrid mode across India.

5. Problem Statement and Industry Context

Recruiters need to screen resumes and determine which candidates are suitable for particular job requirements. A manual process becomes difficult to scale when the number of applications increases. The project therefore explored an AI-assisted approach capable of extracting skills, comparing candidate profiles with job requirements, and ranking suitable roles.

The solution described in the presentation processes resume text, extracts relevant skills and experience, calculates job–candidate similarity, and provides ranked recommendations.

6. Project Objectives

The project was organized around four core objectives:

- Resume Parsing: Extract education, experience, skills, projects, and keywords from resume text.
- Skill Extraction: Identify technical and professional skills using NLP and keyword/semantic matching.
- Job Matching: Compare candidate profiles with job descriptions using vector similarity.
- Recommendation: Rank suitable job roles and provide relevant recommendations.

Together, these objectives form an end-to-end pipeline from raw text input to ranked job recommendations.

7. Technologies and AI Toolchain

The internship presentation identifies the following technology areas:

Programming & Data: Python, Pandas, NumPy, and data preparation.

NLP Processing: NLTK/spacy, text cleaning, tokenization, and extraction.

Machine Learning: Scikit-learn, TF-IDF, embeddings, and similarity scoring.

Application Layer: Streamlit and interactive ranked recommendations.

The project therefore combines conventional text-processing techniques with machine-learning-based similarity calculations and an interactive application layer.

8. System Architecture

The architecture shown in the presentation follows a sequential pipeline:

Resume / Job Input → Text Preprocessing → Feature Extraction → Similarity Model → Ranked Recommendations.

The data layer contains resume documents and job descriptions with structured role metadata. The feature-engineering stage converts text into TF-IDF vectors, semantic representations, and skill features. The recommendation layer calculates similarity, applies ranking rules, and returns matching roles.

9. Methodology and Workflow

The recommendation workflow consists of four major stages.

1. Project Input: The system receives resume text, a target role, candidate skills/experience, or a job description.
2. Feature Analysis: The system extracts skills, education, experience, keywords, and role requirements.
3. AI Matching: NLP features and matching logic are used to generate similarity scores.
4. Output: The system returns ranked job recommendations with match scores and relevant skills.

This workflow provides a structured way to transform unstructured recruitment information into an interpretable recommendation output.

10. Implementation Work

The key project work described in the source presentation included:

Data & Preprocessing: Preparing resume/job text, cleaning noisy content, normalizing terminology, and creating structured inputs.

NLP & Feature Engineering: Applying tokenization, stop-word handling, keyword extraction, and TF-IDF/semantic features.

Similarity & Ranking: Using cosine similarity and weighted matching to rank candidate–job compatibility.

Recommendation Logic: Designing rules to combine skills, experience, education, and role requirements.

Practical Outcome: Developing an end-to-end workflow from raw text input to explainable job recommendations.

11. Internship Timeline

Week 1: Explore the problem, review resume/job datasets, examine text fields, and understand project requirements.

Week 2: Clean resume and job-description text, extract skills, and prepare NLP features.

Week 3: Implement similarity scoring, ranking logic, and the recommendation workflow.

Week 4: Test recommendations, refine the workflow, and complete project documentation.

The timeline reflects progressive development from problem understanding and data preparation to matching, testing, and documentation.

12. Project Performance Indicators

The source presentation includes a project dashboard illustrating an example AI resume-screening output: an 87% resume match score, 14 skills detected, and 5 recommended roles. These values are presented in the internship deck as project/interface indicators rather than as an independently validated benchmark.

The inclusion of these indicators demonstrates how the system can expose understandable outputs to a user: an overall compatibility score, detected skills, and a limited ranked set of recommended roles.

13. Skills Demonstrated and Learning Outcomes

The internship provided practical exposure to data preprocessing, Natural Language Processing, feature engineering, similarity computation, ranking, and recommendation logic. It also demonstrated how these components can be integrated into an application workflow.

Key learning outcomes include:

- Understanding an AI problem in a recruitment context.
- Preparing and normalizing unstructured text data.
- Extracting useful NLP features from resumes and job descriptions.
- Applying TF-IDF and similarity-based matching concepts.
- Designing ranking and recommendation logic.
- Translating an ML workflow into a practical application output.
- Documenting an end-to-end AI project.

14. Advantages of the Proposed System

The project approach can reduce repetitive manual screening effort by automating initial text analysis and matching. It provides a consistent computational method for comparing candidate information with job requirements and can return ranked recommendations instead of an unstructured list.

The modular architecture also makes it possible to improve individual stages—such as preprocessing, feature extraction, similarity modeling, or ranking—without redesigning the entire workflow.

15. Limitations

The source presentation does not provide a formal evaluation dataset, baseline comparison, precision/recall measurements, or detailed validation methodology. Therefore, the project performance should not be interpreted as a rigorously benchmarked recruitment model from the information available in the deck.

A production system would also need careful handling of resume formatting variability, incomplete information, ambiguous skills, changing job requirements, and potential bias in automated ranking.

16. Future Scope

Based on the architecture presented, the system could be extended with richer semantic embeddings, improved skill ontologies, more robust experience extraction, configurable recruiter weighting, explainable recommendation reasons, and systematic evaluation against labeled candidate–job matches.

Future development could also include larger and more diverse datasets, continuous model evaluation, bias and fairness checks, improved handling of synonyms and related skills, and deployment as a scalable recruitment-support application.

17. Conclusion

The Artificial Intelligence Internship at Codec Technologies Pvt. Ltd. provided practical experience in designing an AI-driven text-processing and recommendation workflow. The AI-Powered Resume Screening & Job Recommendation System combined NLP, feature engineering, similarity scoring, ranking, and an application layer to address a practical recruitment problem.

The internship work demonstrated the complete progression from problem definition and data preparation to AI matching, recommendations, testing, and documentation. It strengthened practical understanding of how Artificial Intelligence techniques can be integrated into an end-to-end application.

18. Internship Verification

The source presentation includes internship verification material and an official certificate/offer-letter visual. The verification section identifies the internship title as Artificial Intelligence Internship, the organization as Codec Technologies Pvt. Ltd., the project as AI-Powered Resume Screening & Job Recommendation System, and the internship duration as 08/07/2026 – 08/08/2026. The certificate page also identifies the role as Artificial Intelligence Intern and shows the same duration.

These documents are included in the original presentation as supporting verification material.

Source Note

This report was prepared from the supplied 13-page internship presentation. The presentation contains primarily visual slides, including diagrams and verification images; the report preserves the project terminology and facts visible in those pages.